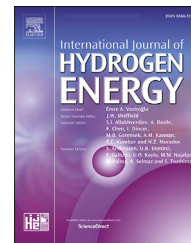


Available online at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/he

Effect of methane reforming before combustion on emission and calorimetric characteristics of its combustion process

Igor Karpilov, Dmitry Pashchenko*

Samara State Technical University, Molodogvardeiskaya 244 St, Samara 443100, Russia

HIGHLIGHTS

- Combustion characteristics of methane/water mixture and products of steam methane reforming were investigated.
- Products of steam methane reforming give higher NO_x emission.
- Increasing in oxidant temperature leads to increasing in NO_x emission.
- Water addition to fuel gives decreasing in NO_x emissions for methane and syngas by 57% and 28%, respectively.
- Maximum flame temperature is higher for syngas at equal combustor heat release.

ARTICLE INFO

Article history:

Received 20 July 2021

Received in revised form

1 September 2021

Accepted 4 September 2021

Available online 24 September 2021

Keywords:

Reforming

Methane

Hydrogen

NO_x emission

Flame

ABSTRACT

In this study, combustion and emission characteristics of methane mixed with steam (CH₄/H₂O) and the products of methane reforming with steam (CO/H₂/H₂O) were compared. Four fuel compositions were analysed: CH₄+H₂O, CH₄+2H₂O, and products of complete methane reforming in these mixtures, respectively. A comparison was carried out through the numerical model created via Ansys Fluent 2019 R2. A combustion process was simulated using a non-premixed combustion model, standard k-ε turbulence model and P-1 radiation model. The combustor heat capacity for interrelated fuel compositions was kept constant due to air preheating before combustion. The inlet air temperature was varied to gain a better insight into the combustion behaviour at elevated temperatures. The effect of steam addition on the emission characteristics and flame temperatures was also evaluated. NO_x formation was assessed on the outlet of the combustion zone. The obtained results indicate that syngas has a higher combustion temperature than methane (in the same combustor heat capacity) and therefore emitted 27% more NO_x comparing to methane combustion. With the air inlet temperature increment, the pollutant concentration difference between the two cases decreased. Steam addition to fuel inlet resulted in lesser emissions both for methane and syngas by 57% and 28%, respectively. In summary, syngas combustion occurred at higher temperature and produced more NO_x emissions in all cases considered.

© 2021 Hydrogen Energy Publications LLC. Published by Elsevier Ltd. All rights reserved.

* Corresponding author.

E-mail address: pashchenkodmitry@mail.ru (D. Pashchenko).

<https://doi.org/10.1016/j.ijhydene.2021.09.043>

0360-3199/© 2021 Hydrogen Energy Publications LLC. Published by Elsevier Ltd. All rights reserved.

Introduction

Hydrocarbon fuels will remain the main source of primary energy in the coming years, despite advances in renewable energy sources. Natural gas is one of the main hydrocarbon fuels. Natural gas is the cleanest burning and fastest-growing fossil fuel, contributing to almost 30% of total energy demand growth through the last decade, more than any other fuel. Being the cleanest burning fossil fuel, natural gas provides a number of environmental benefits compared to other fossil fuels, particularly in terms of air quality and greenhouse gas emissions. However, the combustion of natural gas causes an increase in NO_x emissions.

NO_x emission is one of the main issues for human health and the environment [1]. NO_x forms primarily by the two formation mechanisms: fuel NO and thermal NO. Fuel NO formation is driven by the oxidation of nitrogen-bearing species (HCN, NH_3) that are present in fuel. That mechanism has a small influence if fuel was refined prior to combustion. Authors of this article successfully implemented that technology for the syngas combustion in medium-Btu fueled combustor [2].

Recently, the requirements for NO_x emissions have become more and more stringent. Therefore, scientists and engineers from many countries are working to reduce NO_x emissions [3–5]. Thermal NO is the dominant process, therefore NO_x formation is strongly dependent on the flame temperature [6]. In the scientific articles, a lot of effective ways to reduce NO_x emission were suggested:

- water injection [7–9];
- flue gas recirculation [10,11];
- staged combustion [12];
- low NO_x burners with use of fuel staging, air staging or fiber-matrix, to reduce peak temperature [13,14];
- oxy-fuel combustion [15,16];
- reburning to convert NO to N_2 [17];
- selective catalytic and non-catalytic reduction [18,19].

Among these ways of NO_x emission reduction, water addition to flame is one of the most widely used methods. Water vaporises, thus taking heat from the combustion process, and dilutes combustion products resulting in decreased maximum temperature level and thus NO_x emissions. That technique has some limitations and can be used in systems, where heat loss is tolerable and increased combustion product volume could be beneficial for heat transfer properties.

In recent years, the study of methane dilution with water to reduce NO_x emissions has been of great scientific interest. Solokolou [20] developed a numerical model to analyse the influence of H_2O fuel dilution on the NO_x pollutant formation. 2D premixed combustion model included k- ϵ turbulence model and eddy dissipation model for the turbulence-chemistry interaction. Results showed that with the increment in H_2O diluent concentration, sharp NO_x and slight temperature decrement is observed. The aforementioned effect occurs mostly because of high-temperature region reduction as well as combustion energy absorption by the water diluent. Also, a higher H_2O diluent concentration

influenced the maximum temperature distribution, which was found to shift closer to the inlet.

Dai et al. [21], carried out simulation studies on methane combustion with N_2 and H_2O diluents in hot oxygen coflow. Oxygen concentration in coflow varied from 3% to 85%. Combustion characteristics were influenced by N_2 and H_2O diluents sharper for the lower O_2 concentration. It was observed that combustion occupies a smaller volume and occurs at lower temperatures with H_2O dilution rather than with N_2 dilution. One of the findings is that higher H_2O concentration intensifies CO to CO_2 reaction conversion rate. That reaction is responsible for the major fraction of released heat in the course of methane-air oxidation reaction and therefore combustion volume shrinks. The other reason is that fuel jet entrains more O_2 from the $\text{H}_2\text{O}/\text{O}_2$ hot coflow. The more oxidant is present in the reaction space the higher is combustion reaction rate, that further decreases the reaction volume. Reduced flame temperature is caused by the water vapour higher thermal capacity, which effectively consumes heat of reaction. It also occurred that H_2O diluent has a chemical effect on the combustion reaction, which is expressed by the ignition delay. Ignition delay was observed during MILD combustion. As a result, the flame moved downstream from the fuel nozzle jet. Authors also found out that H_2O diluent has several advantages over CO_2 dilution such as energy efficiency and compact configuration. These advantages are caused mainly by air separation and CO_2 sequestration.

Water addition to the combustion chamber can be organized in different ways. One of those ways is water injection to fuel-air mixture. If we consider a mixing of fuel and water before the combustion chamber, we can expect a fuel reforming reaction. Initial research in the field of fuel reforming before combustion was performed in the United States in 1968. This investigation was conducted by Wimmer and Lee [22]. The main goal of this investigation was to reduce pollutant emissions from spark-ignition engines. Wimmer and Lee were the first who demonstrated that products of the fuel reforming showed better lean burning behaviour and produced lower emissions than propane and methane.

Moreover, in recent years, scientists and engineers are paying more attention to reforming hydrocarbon fuels before combustion by using waste heat [23]. This way of fuel reforming is called thermochemical recuperation (TCR). The main idea of TCR is the endothermic reforming of hydrocarbon fuels to generate hydrogen-rich fuel (syngas or synfuel). Waste heat is energy source for the fuel reforming processes. As a result of this endothermic reforming, a low heating value (LHV) of new synfuel is (in terms of a unit mass of the original fuel) higher than LHV of the original fuel. This new synfuel is used either partially or completely as fuel in such fuel-consuming plants with TCR. In the TCR systems, there is a transformation of flue gases enthalpy into the chemical energy of new synfuel.

One of the most discussed techniques of TCR in the scientific articles is TCR by steam methane reforming [24–27]. The steam methane reforming products mainly consist of hydrogen and carbon monoxide. As mentioned above, one of the most important ways to suppress nitrogen oxides is water injection into the combustion chamber. Fig. 1a shows a

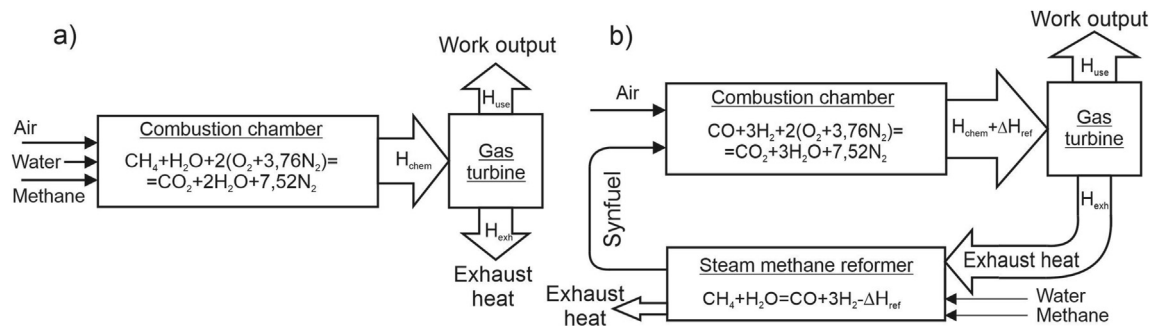


Fig. 1 – A gas turbine with water injection to combustion chamber (a) and with a pre-combustion steam methane reforming (b).

schematic diagram of this method. The main concept is a direct injection of water into the combustion chamber. In fuel-consuming plants with pre-combustion methane reforming organized with the help of the thermochemical waste-heat recuperation system, initially water is used for the methane reforming reaction (Fig. 1b). In other words, the process of fuel combustion in fuel-consuming equipment with TCR is divided into two stages. The first stage is the reforming of methane with steam, the second stage is the combustion of reforming products.

Among the technical problems for an industrial application of the thermochemical waste-heat recuperation systems is investigation of synthesis gas combustion. A distinctive feature of synthesis gas combustion is the presence of hydrogen and carbon monoxide. The development of numerical modelling tools, such as Ansys, Comsol, Star CCM++, OpenFOAM, greatly simplified the study of the emission and calorimetric characteristics of synthesis gas combustion [28]. Norbert Peters showed that up to 70% of combustion problems can be solved using CFD-modeling [29].

Lopez-Ruiz et al. [30] examined NO_x formation during the pure hydrogen combustion process. They investigated the impact of different fuel consumptions on thermal NO_x formation. Case geometry was a 2D axisymmetric model of a domestic multi flame diffusion boiler. The finite-rate chemistry model was implemented in Ansys Fluent 19.0. The authors concluded that NO_x formation could be lowered by 2.2 times at 0.8 kW if the flame-splitting method is used.

Habib et al. [31] carried out simulation studies on emission characteristics of methane and syngas in a package boiler. Combustion of methane and three various syngas compositions were compared. Scientists concluded that boiler exit temperature elevated with the increment in H_2 concentration in syngas due to its slow reaction kinetics. Heat transfer from the flame to the convection part of the burner was higher for the 33% $\text{CO}/67\%\text{H}_2$ fuel which was accompanied by decreased average temperature. Contours of NO_x emissions strongly correlated with maximum temperature. Also, pollutant concentration was decreased by 30% when the amount of excess air was increased from 5% to 25%.

Scribano [32] numerically investigated the influence of different $\text{H}_2:\text{CO}$ ratios, equivalence ratios and Reynolds numbers on CO and NO_x emission characteristics in the process of syn-gas combustion in the radial burner. It was found that the lowest CO and NO_x emissions occurred at the highest

$\text{H}_2:\text{CO}$ fuel ratio. Optimum fuel combustion conditions were observed at a low equivalence ratio and high Reynolds number.

Asgari et al. [33] modeled premixed syngas combustion at different H_2/CO ratio (from 0.25 to 1) and equivalence ratio (from 0.5 to 1) in a McKenna burner. The numerical model was investigated in Open Foam Framework under 2D axisymmetric conditions. It was found that both near flame temperature and NO concentrations decreased with the increment in H_2/CO ratio. Also, authors established that above the flame NO_2 concentration steadily decreases due to a NO_2-NO conversion reaction.

Lee [34] experimentally investigated the impact of flame temperature and flame structure on NO_x and CO emissions under various $\text{CO}/\text{H}_2/\text{CH}_4$ fuel composition in a partially-premixed gas turbine model combustor. Results indicated that NO_x formation is mostly influenced by flame temperature and residence time in a high-temperature region. Residence time is affected by flame structure, which depends on the fuel composition.

Alavandi [35] performed an experimental study to gain better insight into lean-premixed combustion of varying $\text{CH}_4/\text{CO}/\text{H}_2$ mixture in two-section porous burner. Results indicated that NO_x and CO emissions decreased for the higher percentage of CO and H_2 in fuel. However, experiments were conducted at an adiabatic flame temperature below 1600 °C, which corresponds to insignificant thermal NO_x production.

Chaouki [36] compared gas temperature, CO_2 and NO_x emissions for different syngas compositions and methane in a gas turbine. For this purpose, the 3D numerical model was developed consisting of k-e turbulence model, P-1 radiation model and probability density function (PDF) combustion model. Exit average CO_2 emissions per unit of energy generation (kg/kJ) were found lower by 12% only for syngas with high hydrogen content, the other syngas compositions showed higher concentrations than that of methane. Assessing NO_x emissions per unit of energy generation (kg/kJ) the reduction by 33% in comparison to methane combustion was only observed for one syngas composition with high water concentration (22,6%).

Giles et al. [37] analysed the impact of syngas composition and diluents effect on NO_x emission characteristics of syngas non-premixed counterflow combustion. They developed a numerical model using the OPPDIF code with the Chemkin package. Authors found that H_2O is the most efficient diluent

comparing to CO₂ and N₂ due to its high specific heat which efficiently lowers the flame temperature. Also, it was observed that the thermal mechanism is dominant in NO_x formation for 50%H₂:50%CO mixture.

Such a variety of articles on CFD-modeling of synthesis gas combustion is due to the fact that the composition of synthesis gas can vary over a wide range of component concentrations. In addition, the development of the thermochemical waste-heat recuperation technology and the on-board hydrogen production technology makes it extremely relevant to answer the question: “How does the reforming of methane affect the emission and calorimetric characteristics of its combustion process?” In this article, using CFD-modeling, we tried to show the effect of methane reforming on the emission and calorimetric characteristics of its combustion process.

Numerical setup

Original burner geometry was simplified to a 2D case and considered 2D axisymmetric. In our previous research [38] we showed that the geometric dimensionality has an impact on the results of numerical simulations at a low L/D ratio (less than 5). Therefore, in this study, we chose a L/D ratio of 7.6. Thus, the dimension of the geometry will not have a noticeable effect on the results of numerical simulations. Basic burner geometry contains separated coaxial inflow tubes for air and fuel supply. The symmetry axis is located in the combustor centre. Therefore only half of the air inlet with inner and outer diameter of 22,2 mm and 57,2 mm, respectively and half of the fuel inlet with a diameter of 16 mm are presented. Computational mesh consisted of uniform elements with quadrilateral form, the total number of elements is 245120. This mesh structure was chosen according to the findings in authors' previous investigations [38,39]. Computational domain and mesh structure can be seen in Fig. 2.

Governing equations

Ansys Fluent software was utilised for numerical investigation. Fluent solves the governing equations for the mass and momentum conservation, energy and other scalars such as chemical species and turbulence. In both cases, a control-volume-based technique is used.

The equation for conservation of mass or continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 \quad (1)$$

where ρ is the flow density, kg/m³, $\vec{v}$ is gas velocity, m/s.

In the 2D axisymmetric case form of the expression changes:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho v_x) + \frac{\partial}{\partial r}(\rho v_r) + \frac{\rho v_r}{r} = 0 \quad (2)$$

where x is the axial coordinate, r is the radial coordinate, v_x is the axial velocity and v_r is the radial velocity.

Conservation of momentum in an internal (non-accelerating) momentum equation for 3D geometries:

$$\nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla p + \nabla \cdot (\vec{\tau}) + \rho \vec{g} + \vec{F} \quad (3)$$

where p is the static pressure, $\vec{\tau}$ is the stress tensor, $\rho \vec{g}$ is the gravitational body force and $\vec{F}$ is external body force.

For 2D axisymmetric geometries the axial and radial momentum equations could be found in Ref. [40].

Stress tensor is defined by the following equation:

$$\vec{\tau} = \mu \left[\left(\nabla \vec{v} + \nabla \vec{v}^T \right) - \frac{2}{3} \nabla \cdot \vec{v} \vec{I} \right] \quad (4)$$

where μ is the molecular viscosity, $\vec{I}$ is the unit tensor and $\nabla \vec{v} + \nabla \vec{v}^T$ is the effect of volume dilation.

The standard k- ϵ turbulence model was chosen for the turbulence mixing modelling according to numerous authors [36,41–44]. This technique offers good convergence and isn't memory intensive. Following transport equations describe the turbulence kinetic energy k and ϵ - its rate of dissipation:

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + G_k + G_b - \rho \epsilon - Y_M + S_k \quad (5)$$

$$\begin{aligned} \frac{\partial}{\partial t}(\rho \epsilon) + \frac{\partial}{\partial x_i}(\rho \epsilon u_i) = & \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x_j} \right] + C_{1\epsilon} \frac{\epsilon}{k} (G_k + C_{3\epsilon} G_b) \\ & - C_{2\epsilon} \rho \frac{\epsilon^2}{k} + S_\epsilon \end{aligned} \quad (6)$$

In the equations above G_k is a term, that defines the generation of turbulent kinetic energy due to the mean velocity gradients. G_b represents the generation of turbulence kinetic energy due to buoyancy. Y_M is responsible for the contribution of the fluctuating dilatation in compressible turbulence to the overall dissipation rate. $C_{1\epsilon}$, $C_{2\epsilon}$ and $C_{3\epsilon}$ are constants, σ_k and σ_ϵ are the turbulent Prandtl numbers for k and ϵ , respectively, S_k and S_ϵ are user-defined source terms.

For the non-premixed non-adiabatic combustion model the total enthalpy form of energy equation is solved [40]:

$$\nabla \cdot (\rho \vec{v} H) = \nabla \cdot \left(\frac{k_t}{c_p} \nabla H \right) + S_h \quad (7)$$

where S_h includes the heat of chemical reaction and any other defined volumetric heat source, k_t is the turbulent thermal conductivity, H is the total enthalpy and c_p is specific heat of the mixture.

The total enthalpy H is defined by the expression:

$$H = \sum_j Y_j H_j \quad (8)$$

where Y_j is the mass fraction of species j and H_j is the term described by:

$$H_j = \int_{T_{ref}}^T c_{p,j} dT + h_j^0(T_{ref,j}) \quad (9)$$

where $h_j^0(T_{ref,j})$ is the formation enthalpy of species j at the reference temperature $T_{ref,j}$.

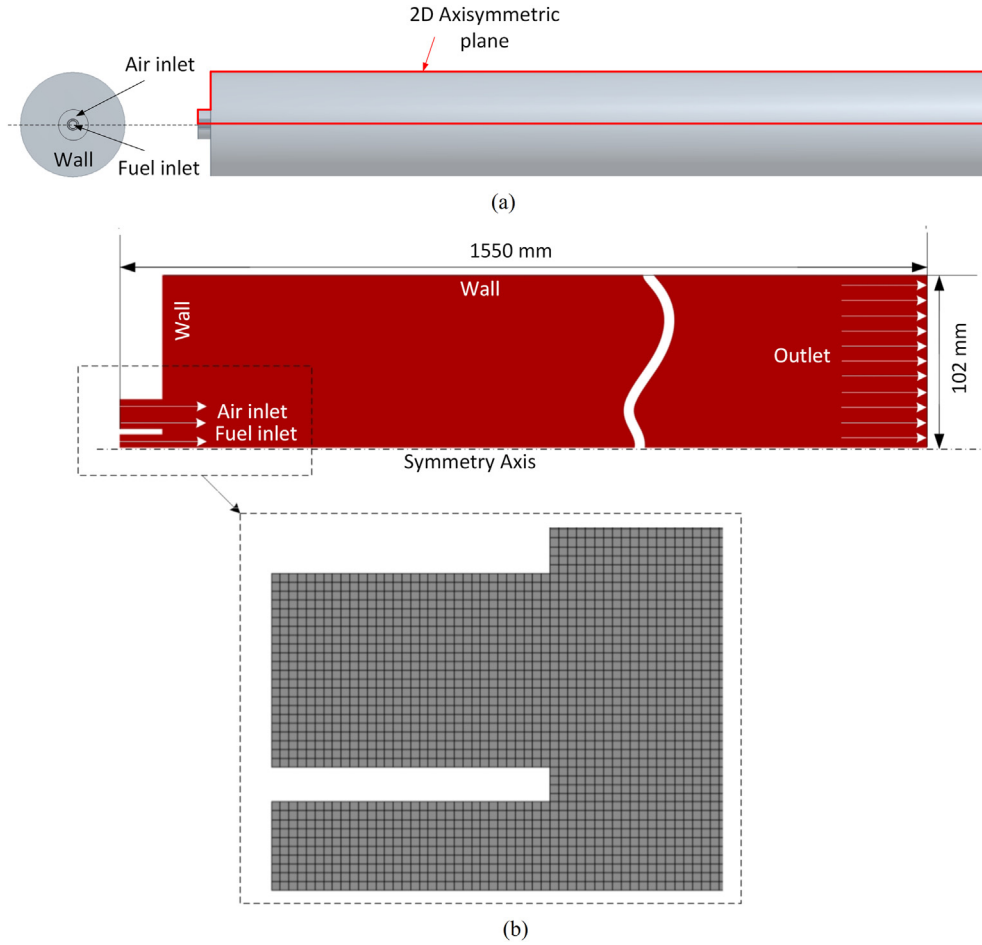


Fig. 2 – (a) 3D burner geometry (b) 2D axisymmetric geometry with mesh structure.

As the combustion process takes place at high temperatures, the one of main heat transfer process is radiation. For the valid temperature distribution in the combustor chamber, a proper radiation model should be implemented. For the current case, the P_1 radiation model was chosen. This model is the simplest case of the general P–N model. The gray radiation was modeled by the following equation:

$$\nabla \cdot (\Gamma \nabla G) + 4\pi \left(an^2 \frac{\sigma T^4}{\pi} + E_p \right) - (a + a_p)G = 0 \quad (10)$$

where G is the incident radiation, a is the absorption coefficient, n is the refractive index of the medium, σ is the Stefan-Boltzmann constant, E_p is the equivalent emission of the particles and a_p is the equivalent absorption coefficient.

The parameter Γ in eq.(10) is defined as:

$$\Gamma = \frac{1}{(3(a + \sigma_s) - C\sigma_s)} \quad (11)$$

where σ_s is the scattering coefficient and C is the linear-anisotropic phase function coefficient.

Equivalent emission of the particles is written in form:

$$E_p = \lim_{V \rightarrow 0} \sum_{n=1}^N \epsilon_{pn} A_{pn} \frac{\sigma T_{pn}^4}{\pi V} \quad (12)$$

where ϵ_{pn} , A_{pn} and T_{pn} are the emissivity, projected area and temperature of particle n . The summation is over N particles in volume V .

Equivalent absorption coefficient is described as following:

$$a_p = \lim_{V \rightarrow 0} \sum_{n=1}^N \epsilon_{pn} \frac{A_{pn}}{V} \quad (13)$$

For absorption coefficient the Weighted-Sum-of-Gray-Gases Model (WGGSM) was implemented. WGGSM model proposes a good trade-off between the oversimplified gray gas model and a complete model which considers particular absorption bands. Detailed description of that modeled could be found in Ref. [45].

Combustion model

It is known that for non-premixed combustion modeling, transport equations for one or two conserved scalars (the mixture fractions) are solved. Since the equations for individual species are not solved, the predicted mixture fraction fields determine species concentrations. The β function - Probability Density Function (PDF) accounts for the turbulence-chemistry interaction [46].

Based on the assumption that the instantaneous thermochemical state of the fluid is related to mixture fraction (a

conserved scalar quantity) under certain set of simplifying assumptions the non-premixed combustion can be modeled. The mixture fraction f in terms of the atomic mass fraction can be determined based on the following expression [47]:

$$f = \frac{Z_i - Z_{i,ox}}{Z_{i,fuel} - Z_{i,ox}} \quad (14)$$

Z_i - the elemental mass fraction of element, i ; the subscript ox represents the value at the oxidizer stream inlet and the subscript $fuel$ represents the value at the fuel stream inlet.

Under the assumption of equal diffusivities, the species equations can be reduced to a single equation for the mixture fraction, f . The reaction source terms in the species equations cancel (since elements are conserved in chemical reactions), and therefore is a conserved quantity. While the assumption of equal diffusivities is problematic for laminar flows, it is generally acceptable for turbulent flows where turbulent convection overwhelms molecular diffusion. The Favre mean (density-averaged) mixture fraction equation is [48]:

$$\frac{\partial}{\partial t}(\rho \bar{f}) + \nabla \cdot (\rho \bar{v} \bar{f}) = \nabla \cdot \left(\left(\frac{\mu_t}{\sigma_t} + \frac{k}{C_p} \right) \nabla \bar{f} \right) + S_m + S_{user} \quad (15)$$

where k is mixture laminar thermal conductivity, C_p is the specific heat of mixture, σ_t is the Prandtl number, and μ_t is the turbulent viscosity. S_m is the source term that indicates transfer of mass of liquid fuel droplets or reacting particles into the gas phase; S_{user} is the user defined source term.

A conservation equation for the mixture fraction variance is solved in addition to solving for the Favre mean mixture fraction:

$$\begin{aligned} \frac{\partial}{\partial t}(\rho \overline{f'^2}) + \nabla \cdot (\rho \bar{v} \overline{f'^2}) = \nabla \cdot \left(\left(\frac{\mu_t}{\sigma_t} + \frac{k}{C_p} \right) \nabla \overline{f'^2} \right) \\ + C_g \mu_t (\nabla \bar{f})^2 - C_d \rho \frac{\epsilon}{k} \overline{f'^2} + S_{user} \end{aligned} \quad (16)$$

where $f' = f - \bar{f}$, C_g and C_d are the constant values. Turbulence-chemistry interactions in the closure model are described by the mixture fraction variance.

The main advantage of mixture fraction modelling approach is the chemistry reduction to one or two conserved mixture fractions. Species fractions, density and temperature are uniquely related to the mixture fractions with the chemical equilibrium assumption [40].

In this study, the NO_x formation was evaluated by the formation of NO molecules. That is a correct assumption for most cases as N_2O and NO_2 formation percentage is quite low [42]. NO formation was only considered through thermal and prompt pathways. Fuel formation could be neglected due to the assumption, that no nitrogen molecules are present in the inlet fuel stream. As was shown by authors of new article [49], fuel and intermediate formation processes do not have a significant impact on the overall pollutant concentration.

Mass conservation equation of NO species is solved considering the major mixture properties: convection, diffusion, production, and consumption of NO and related species. It can be written as:

$$\frac{\partial}{\partial t}(\rho Y_{\text{NO}}) + \nabla \cdot (\rho \bar{v} Y_{\text{NO}}) = \nabla \cdot (\rho D \nabla Y_{\text{NO}}) + S_{\text{NO}} \quad (17)$$

where Y_{NO} is the mass fraction of NO in the gas phase; D is the effective diffusion coefficient; S_{NO} is the NO source term.

Thermal NO_x formation was modeled using the extended Zeldovich mechanism. That mechanism contains 3 separate highly temperature dependent chemical reactions:



Under stoichiometric conditions it can be assumed that the rate of formation and consumption of N atoms is equal. Then the NO formation rate is described by:

$$\frac{d[\text{NO}]_{\text{thermal}}}{dt} = 2k_{f,1}[\text{O}][\text{N}_2] \frac{\left(1 - \frac{k_{r,1}k_{r,2}[\text{NO}]^2}{k_{f,1}[\text{N}_2]k_{f,2}[\text{O}_2]}\right)}{\left(1 + \frac{k_{r,1}[\text{NO}]}{k_{f,2}[\text{O}_2] + k_{f,3}[\text{OH}]}\right)} \quad (21)$$

where $k_{f,1} = 1,8 \cdot 10^8 e^{-38370/T}$; $k_{f,2} = 1,8 \cdot 10^4 e^{-4680/T}$; $k_{f,3} = 7,1 \cdot 10^7 e^{-450/T}$; $k_{r,1} = 3,8 \cdot 10^7 e^{-425/T}$; $k_{r,2} = 3,81 \cdot 10^3 e^{-20820/T}$ - are the rate constants for these reactions, obtained from the experiments [50].

The kinetics of thermal NO_x formation occurs generally much slower than the combustion reaction kinetics. Therefore NO_x formation could be calculated when the combustion process reaches equilibrium state [51]. Hence the equilibrium or, like in this case, the modified partial-equilibrium approach could be used for the calculation of O-atom and OH radical molar concentration:

$$[\text{O}] = 36.64 T^{1/2} [\text{O}_2]^{1/2} e^{-27123/T} \quad (22)$$

$$[\text{OH}] = 2.129 \times 10^2 T^{-0.57} e^{-4595/T} [\text{O}]^{1/2} [\text{H}_2\text{O}]^{1/2} \quad (23)$$

With the derived O and OH molar concentration, the predicted NO source term in the result of thermal formation mechanism is:

$$S_{\text{NO, thermal}} = M_{\text{NO}} \frac{d[\text{NO}]_{\text{thermal}}}{dt} \quad (24)$$

where M_{NO} is the molecular weight of NO.

The second NO_x formation mechanism - prompt NO_x occurs under low temperatures in fuel-rich conditions with short residence time. It is evident, that these regions form in combustion environments similar to the one presented in the current study [52]. The mechanism involves a chain of reactions and intermediate species:



The early stage of combustion takes place in a fuel-rich environment. The prompt NO_x formation mechanism is dominant in that zone as the temperatures are not high enough for thermal formation mechanism and due to the high oxygen

concentration, the NO forms practically exclusively rather than N radical. For the current numerical modelling De Soete kinetic model, improved by Backmier experimental observations, was utilised. Prompt NO_x reaction rate is therefore described as:

$$\frac{d[\text{NO}]_{\text{prompt}}}{dt} = \varphi \cdot 6,4 \cdot 10^6 \left(\frac{RT}{p} \right)^2 [\text{FUEL}] [\text{O}_2]^b [\text{N}_2] e^{-E_a/RT} \quad (29)$$

where φ is the correction factor, thoroughly described in Ref. [40]; b is the oxygen reaction order; E_a is reaction activation energy.

And the source term for prompt mechanism is expressed:

$$S_{\text{NO, prompt}} = M_{\text{NO}} \frac{d[\text{NO}]_{\text{prompt}}}{dt} \quad (30)$$

Boundary conditions

The numerical model was computed via Ansys Fluent R2. Standard k- ϵ turbulence model with Scalable wall function was chosen for turbulence interaction due to its economy and reliability with reasonable accuracy. The pressure-based Navier-Stokes solution algorithm was utilised with Simple pressure-velocity coupling method. Calculations were kept for all cases under constant thermal power. The main operating conditions: mixture composition, fuel mass flow and excess air ratio can be seen in Table 1. Inlet mass flow of methane was less than that of syngas due to the fact, that the mass of syngas generated in steam methane reforming was 2125 times higher than the initial methane mass before reforming. Mass variance results in enthalpy difference which was neutralised by the increment in air inlet temperature. Inlet air temperature varied from 515 °C to 1082 °C for methane and from 27 °C to 627 °C for syngas combustion. Fuel inlet temperature was given at 274 °C and kept constant for all cases. Pressure outlet boundary condition was utilised with backflow specified temperature at 1527 °C. Standard discretisation scheme used for pressure equation. Second-order upwind interpolation was employed for the above-mentioned equations. Convergence criteria were set to 10^8 for energy, 10^6 for pollutant NO and radiation and 10^3 for other equations. The solution was calculated until mentioned convergence criteria were met.

Results

For the combustion performance comparison of the initial and thermochemically generated fuel, namely methane and syngas, case 1 and case 3 with varying inlet air temperatures were studied and discussed below. Case 1 represents methane combustion with water additions. The mass fractions of methane and water were chosen accordingly to the fuel

required for the thermochemical recuperation process, which corresponds to the right part of steam methane reforming reaction (Fig. 1). Syngas in case 3 consists of H₂ and CO, formed as products of reforming of water and methane presented in case 1 and representing the left part of steam methane reforming reaction (Fig. 1). As seen in Table 1, the fuel inlet mass flow is kept the same, while stoichiometric air mass flow is also almost equal. So it can be noted, that the same amount of fuel and air is burnt in both cases, but the mixture of CO and H₂ has a higher low heating value than that of CH₄ and H₂O mixture. The correct comparison of pollutant formation requires the same heat generation value. Thus, the enthalpy difference should be addressed. For the enthalpy difference compensation, the inlet air temperature was varied. In case 1 air flow temperature was elevated until the same heating value was achieved as in case 3. It is widely known that air specific enthalpy depends on its temperature [53]. Therefore, the enthalpy calculator available by the following link [54] was utilised. Inlet fuel temperature was chosen to be 274 °C for both cases according to the water vapour parameters.

Temperature contours for cases 1 and 3 at air inlet temperature 515 °C and 27 °C correspondingly are shown in Fig. 3. It can be seen, that the whole flame structure is generally the same due to the equal fuel and air mass flow rate. High outlet temperature and temperature uniform distribution close to the outlet are due to the wall adiabatic boundary condition. That boundary condition is encountered in the combustion chamber of gas turbine units, where wall heat loss is insignificant [55]. Syngas combustion takes place at a higher temperature, whereas the average outlet temperature for both cases is practically the same. That effect is explained by the different flame temperatures: 1937 °C, 2127 °C and 2127 °C for methane, hydrogen and carbon monoxide, respectively [56]. The distinction is intensified by methane dilution with vapour, which leads to even lesser flame temperature.

Pollutant distribution in the combustor in cases 1 and 3 with air inlet temperatures 27 °C and 515 °C respectively could be seen in Fig. 4. The main formation zone is placed near the region with maximum temperature. That occurs due to the dominant thermal NO formation mechanism. In addition to that mechanism, residence time influences NO formation as well [57]. Thus, the formation zone is shifted away from the maximum temperature zone. Syngas burns at a higher temperature, therefore the pollutant forms very intensively in case 3. The aforementioned water dilution in case 1 further enlarges discrepancies. After formation pollutant blends homogeneously with exhaust gases. For case 3 the numerical values are not obvious from the presented contour, so they are presented below.

As in case 1 water dilution was used, the decision was made to investigate combustion behaviour with more water.

Table 1 – Boundary conditions for fuel compositions.

Case number	CH ₄ (% by mass)	H ₂ (% by mass)	CO (% by mass)	H ₂ O (% by mass)	Fuel mass flow (kg/s)	Air mass flow (kg/s)	Excess air ratio
Case 1	47.1	00.0	00.0	52.9	0.000425	0.00378	1
Case 2	30.8	00.0	00.0	69.2	0.000650	0.00378	1
Case 3	00.0	17.8	82.2	00.0	0.000425	0.00383	1
Case 4	00.0	11.6	53.8	34.6	0.000650	0.00383	1



Fig. 3 – Temperature distribution in combustor chamber at air temperature 27 °C and 515 °C, where upper and bottom contours represent Case 3 and Case 1, respectively.

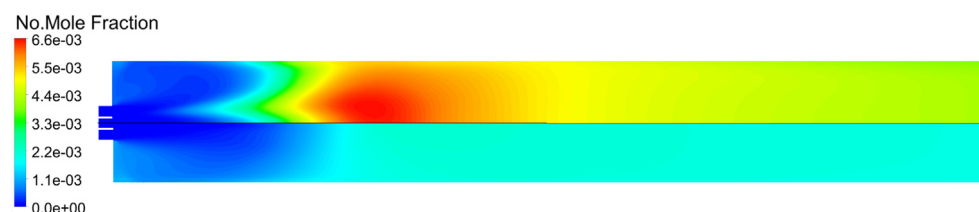


Fig. 4 – NO_x mole fraction distribution in combustor chamber at air temperature 27 °C and 515 °C, where upper and bottom contours represent Case 3 and Case 1, respectively.

The amount of water added to the fuel inlet is 1 mol of H₂O per mole of fuel. As a result, case 2 and case 4 differ from case 1 and case 3 by the same mass of water: 2, 25e⁻⁴ kg/s, respectively. That is meant to understand the effect of water vapour addition on the syngas combustion, as well as to compare the results with case 1 and case 3. Fig. 5 illustrates temperature contours for cases 4 and 2 under 515 °C and 27 °C inlet air temperature, respectively. It can be seen, that the flame temperature decreases but the overall flame structure is generally the same as in case 1 and case 3. Also, the later ignition in case 4 could be noted, as the peak temperature is further downstream. This is attributed to the reaction rate, which is higher for the H₂ combustion than that of methane [58]. Nevertheless, the main reaction zone for both cases extends similarly. That effect could be attributed to the slower combustion of CO, which requires higher residence time [31].

Comparison of pollutant formation in case 2 and case 4 is presented in Fig. 6. Although the air inlet temperatures are the same as in case 1 and case 3, the maximum NO concentration is decreased practically by two times. The difference is due to the lesser temperature as a result of water vapour addition. The main NO formation zone in syngas combustion does not increase in size, while the overall mixing with the combustion products intensifies. Numerical values for methane and syngas are presented below.

Numerical values for pollutant formation in all cases under varying air inlet temperatures are given in Fig. 7. Pollutant

concentration is written in ppm and two scales for temperature are presented. The upper-scale defines methane air temperature and the bottom - syngas inlet air temperature. The reason for implementing the two-scale system is the enthalpy difference between methane + water and syngas mixtures. That difference is caused by the various lower heating value of the two fuels. To compensate for the lacking heat, methane oxidizer (air) was preheated, consequently increasing its enthalpy. As a result, the second scale is needed to present the methane inlet air temperature. With air temperature increment, the rise in maximum flame temperature is observed. Due to higher temperature, NO emission formation intensifies, as it has a very strong dependence on the combustion temperature [59]. On the other side, dilution with water significantly lowers flame temperature and as a result NO formation. That behaviour is characteristic of the cases 2 and 4, where extra water was added. Moreover, for all cases it is observed, that pollutant concentration and maximum temperature have similar dependencies. The nature of trend, depicted in Fig. 7 strongly correlates with data found in the articles [60,61]. Generally, syngas is considered a cleaner fuel than methane, but from the pollutant formation perspective, it can be seen that under similar operating conditions CH₄ emits lesser NO particles. The main reason behind this is higher flame temperatures and longer residence time in the course of syngas combustion.

Axial and radial temperature distribution analysis helps to gain a better insight into the combustion dynamics. The

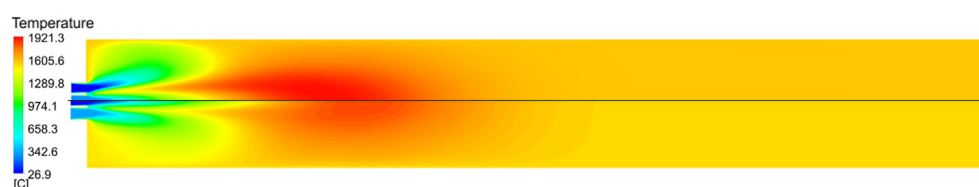


Fig. 5 – Temperature distribution in combustor chamber at air temperature 515 °C and 27 °C, where upper and bottom contours represent Case 4 and Case 2, respectively.



Fig. 6 – NO_x mole fraction distribution in combustor chamber at air temperature 27°C and 515°C , where upper and bottom contours represent Case 4 and Case 2, respectively.

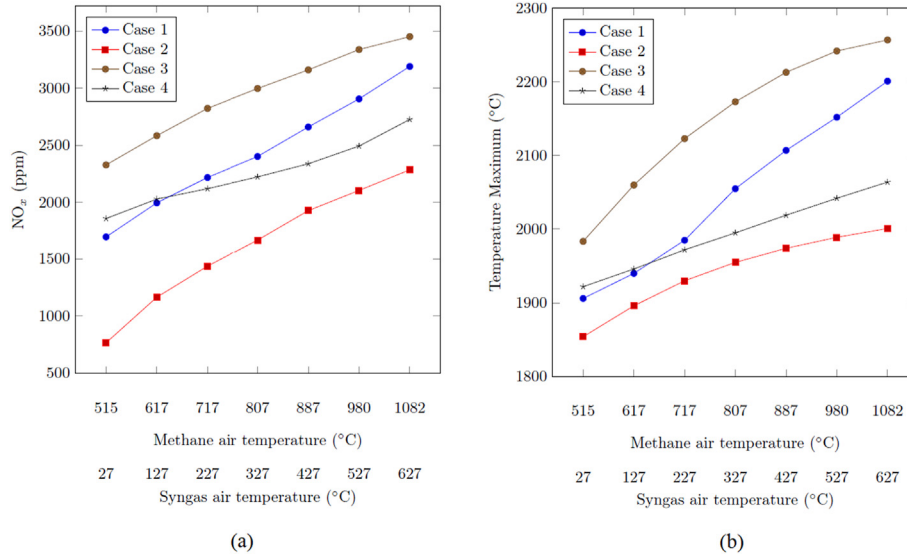


Fig. 7 – (a) NO_x average outlet concentration and (b) Temperature maximum in the combustion zone is presented under different inlet air temperatures for each pair of cases.

values calculated at lowest air inlet temperatures are presented in Fig. 8. Axial values show the temperature distribution across the centerline of the combustor. To study the temperature values in the radial direction were used averaged values across the radial coordinate (from $r = 0$ to $r = 0.102$ m) for a given axial coordinate (x). From the axial temperature distribution it could be deduced, that the combustion for both cases has a similar form, but different

maximum temperature distributions. Syngas fuel starts to burn earlier than methane due to the lower flammability limit and also has longer flame length due to the faster gas velocities as H_2 density is lower than that of CH_4 [49]. For both case 3 and 4 the first values of average radial temperature distribution are rather high, then significantly lower and after that rise steadily. The observed effect is attributed to the heated wall near the burner for the first values, then

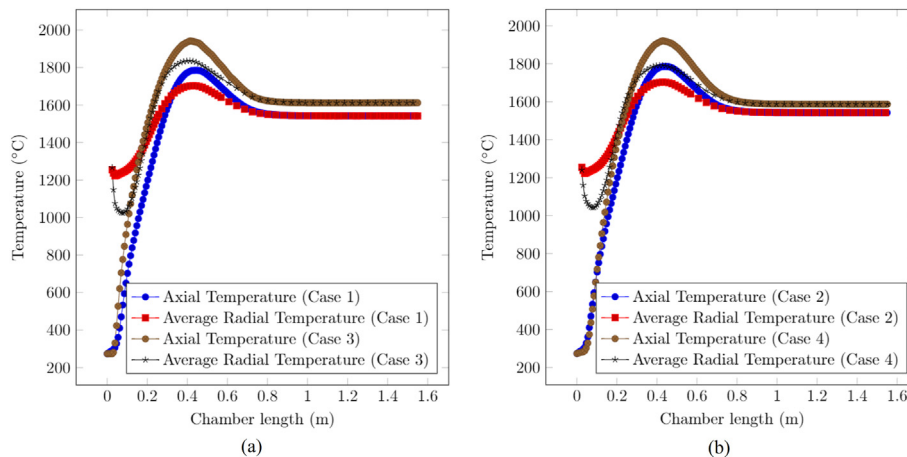


Fig. 8 – Axial temperature across the centerline of the combustor ($r = 0$) and averaged radial temperature at different chamber length in (a) case 1 and case 3 and (b) case 2 and case 4 at 515°C and 27°C air inlet temperature, respectively.

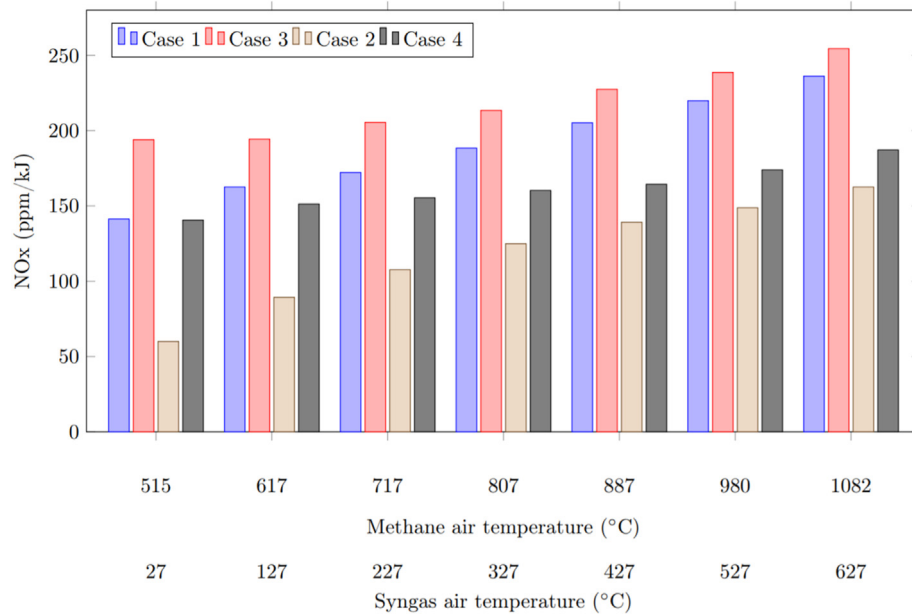


Fig. 9 – Average outlet pollutant concentration per unit of generated heat for varying inlet air temperature for each pair of cases.

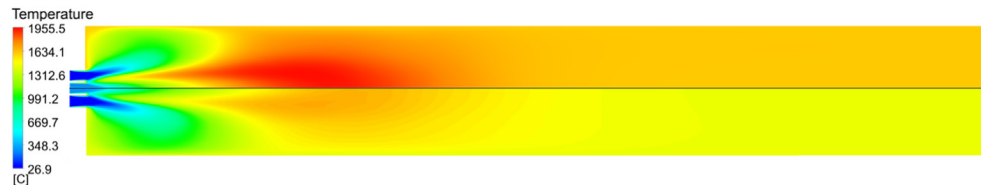


Fig. 10 – Temperature distribution in combustor chamber at equal inlet air temperature 27 °C, where upper and bottom contours represent Case 3 and Case 1, respectively.

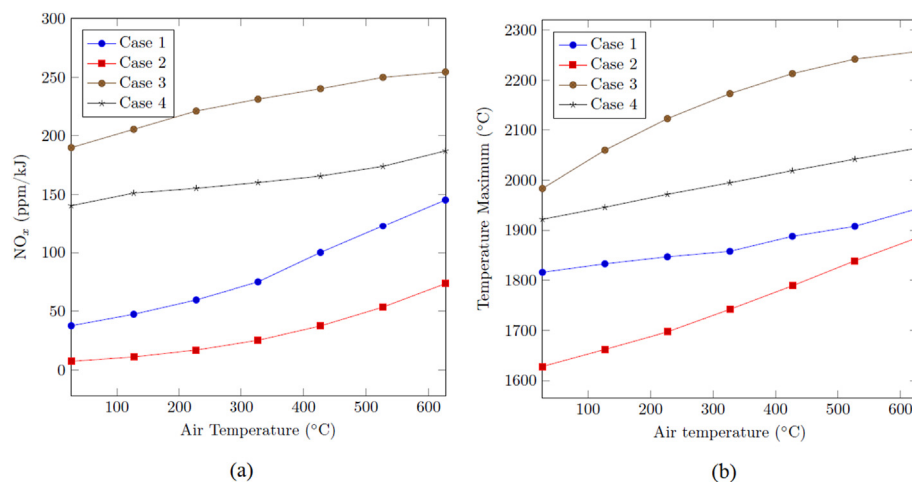


Fig. 11 – (a) NO_x average outlet concentration per unit of generated heat and (b) Combustion temperature maximum are shown for the same air inlet temperature in all cases.

the cold air and fuel mixing which produces lower radial temperatures and then combustion occurs with significant heat release. In case 1 and 2 preheated air was used, so the mean radial temperature reduction at the section near the combustor wall is insignificant.

The comparison of average outlet pollutant concentration per unit of generated heat for syngas and methane combustion with and without water dilution under different inlet air temperatures is provided in Fig. 9. It is established that in flue gases pollutant concentration is significantly higher for the

cases with the same fuel but without water addition. Water has a rather high heat capacity and therefore the ability for temperature and thermal NO pollutant reduction. Combustion of 30,8% methane and 69,2% H₂O mixture resulted in the least emissions, while the pure syngas combustion emitted the most. The results are consistent with the findings of Büyükakin [49]. For all cases, a similar NO growth rate is observed, however in case 2 for inlet air temperatures at 515 °C, 617 °C and 717 °C it was found faster than for other cases. Another interesting tendency is the faster growth rate of NO per unit of heat generation for methane burning mixtures. This dependence leads to a conclusion, that methane combustion is more favourable with low inlet air temperature, and syngas combustion emits similar pollutant quantities with high oxidizer temperature. For example, average outlet pollutant concentration difference in case 1 and case 3 is 37,5% and 8% for minimum and maximum temperature, respectively. Case 2 and case 4 demonstrate similar dependence only with higher amplitude of 134% and 15%.

In this section simulation results without air preheating are presented. Therefore, the supplied heat was different for methane and syngas combustion. That is meant to underline the importance of the equal enthalpy approach. In Fig. 10 it is clearly observed that temperatures, obtained in the course of methane combustion, are considerably lower. Numerical values, presented in Fig. 11, differ from the values with preheated inlet air by 2 times at average. Average outlet temperatures are also varying greatly, the difference of 178 °C was indicated. These results indicate the correctness of the chosen approach as pollutant formation can not be authentically evaluated under these conditions due to major heat release differences.

Conclusion

In the present study combustion characteristics of unprocessed fuel - a mixture of methane and water and the products of steam methane reforming (syn-gas) were simulated via Ansys Fluent 2019 R2. Syn-gas contained 25% CO and 75% CH₄ by volume and was obtained prior to combustion during the thermochemical recuperation process of methane and water. Equal fuel mass flow and heat release were considered for both cases. With air preheating for methane combustion, equal heat release was achieved. Key findings are:

- The numerical results show that the flame structure is similar for both cases, but combustion temperature is higher for syngas combustion. Due to higher temperature, thermal NO_x formation was intensified, which resulted in a higher average outlet pollutant concentration. Syngas combustion emitted 27% more NO particles than methane combustion at low inlet air temperature. Therefore under similar conditions syngas combustion produced more NO particles.
- The addition of 53% H₂O by mass to fuel inlet at lower air temperature decreased the pollutant formation by 28% and 57% for syngas and methane, respectively. Also, the NO_x formation zone was found larger and distributed more homogeneously throughout the combustor length.

- Obtained values for methane combustion without the initial air heating differed significantly. That demonstrated the necessity of the chosen approach.
- Presented study was conducted in a rather simple burner geometry without any pollutant decreasing techniques except for water addition. The latest findings suggest that syngas rather than methane combustion has more chances of becoming cleaner technology. Syngas combustion performance with other reducing techniques may be investigated in further studies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work is supported by the Russian Science Foundation under grant 19-19-00327.

REFERENCES

- [1] Skalska Kinga, Miller Jacek S, Ledakowicz Stanislaw. Trends in nox abatement: a review. *Sci Total Environ* 2010;408(19):3976–89.
- [2] Hasegawa Takeharu, Tamaru Takashi. Gas turbine combustion technology reducing both fuel-no x and thermal-no x emissions for oxygen-blown igcc with hot/dry synthetic gas cleanup. 2007.
- [3] Wang Yang, Gu Mingyan, Zhu Yuhan, Cao Ling, Zhu Bencheng, Wu Jiajia, et al. A review of the effects of hydrogen, carbon dioxide, and water vapor addition on soot formation in hydrocarbon flames. *Int J Hydrogen Energy* 2021;46(61):31400–27.
- [4] Reale Fabrizio, Sannino Raniero. Water and steam injection in micro gas turbine supplied by hydrogen enriched fuels: numerical investigation and performance analysis. *Int J Hydrogen Energy* 2021;46(47):24366–81.
- [5] Zhang Meng, An Zhenhua, Wang Liang, Wei Xutao, Jianyihan Bieerlan, Wang Jinhua, et al. The regulation effect of methane and hydrogen on the emission characteristics of ammonia/air combustion in a model combustor. *Int J Hydrogen Energy* 2021;46(40):21013–25.
- [6] Kevin J Whitty, Zhang Hongzhi R, Eddings Eric G. Emissions from syngas combustion. *Combust Sci Technol* 2008;180(6):1117–36.
- [7] Anufriev IS. Review of water/steam addition in liquid-fuel combustion systems for nox reduction: waste-to-energy trends. *Renew Sustain Energy Rev* 2021;138:110665.
- [8] Ariemma GB, Sabia P, Sorrentino G, Bozza P, De Joannon M, Ragucci R. Influence of water addition on mild ammonia combustion performances and emissions. *Proc Combust Inst* 2021;38(4):5147–54.
- [9] Lyu Yajin, Aspden AJ, Zhang Linyao, Liu Li, Qiu Penghua. Study of the chemical effect of steam dilution on no formation in laminar premixed h₂/air flame at normal and

- elevated pressure. *Int J Hydrogen Energy* 2021;46(24):13402–12.
- [10] Kim Seong-il, Lee Hosoo, Lee Yong woon, Yang Won. Performance evaluation of a pressurized oxy-combustion power plant according to wet and dry flue gas recirculation. *International Journal of Greenhouse Gas Control* 2021;107:103277.
 - [11] Lee Woo-Sung, Kang Jun-Ho, Lee Jae-Cheol, Lee Chang-Ha. Enhancement of energy efficiency by exhaust gas recirculation with oxygen-rich combustion in a natural gas combined cycle with a carbon capture process. *Energy* 2020;200:117586.
 - [12] Boulahlib MS, Medaerts F, MA Boukhalfa. Experimental study of a domestic boiler using hydrogen methane blend and fuel-rich staged combustion. *Int J Hydrogen Energy* 2021. <https://doi.org/10.1016/j.ijhydene.2021.01.103>.
 - [13] Liang Zhanwei, Chen Hongwei, Zhao Bin, Jia Jiandong, Cheng Kai. Synergetic effects of firing gases/coal blends and adopting deep air staging on combustion characteristics. *Appl Energy* 2018;228:499–511.
 - [14] Jin Yan, Lu Xiaofeng, Zheng Xiong, Liu Zhuo, Fan Xuchen. Enhanced combustion behavior and nox reduction performance in a cfb combustor by combining flue gas recirculation with air-staging: effect of injection position. *J Energy Inst* 2021;96:294–309.
 - [15] Kim Donghee, Yang Won, Kang Y Huh, Lee Youngjae. Demonstration of 0.1 mwth pilot-scale pressurized oxy-fuel combustion for unpurified natural gas without co₂ dilution. *Energy* 2021;223:120021.
 - [16] Gaber Christian, Schluckner Christoph, Wachter Philipp, Demuth Martin, Hochenauer Christoph. Experimental study on the influence of the nitrogen concentration in the oxidizer on nox and co emissions during the oxy-fuel combustion of natural gas. *Energy* 2021;214:118905.
 - [17] Tayyeb Javed M, Irfan Naseem, Gibbs BM. Control of combustion-generated nitrogen oxides by selective non-catalytic reduction. *J Environ Manag* 2007;83(3):251–89.
 - [18] Warnatz Jurgen, Maas Ulrich, Dibble Robert W, Warnatz J. *Combustion*. Springer; 2006.
 - [19] Lei Zhiping, Yan Jingchong, Jia Fang, Shui Hengfu, Ren Shibiao, Wang Zhicai, Li Zhanku, Kong Ying, Kang Shigang. Catalytic combustion of coke and no reduction in-situ under the action of fe–cao and fe–ceo₂. *Energy* 2021;216:119246.
 - [20] Nasiri Soloklou Mohsen, Akbar Golneshan Ali. Numerical study of the effect of h₂o diluents on nox and co formation in turbulent premixed methane-air flame. *Int J Hydrogen Energy* 2020;45(18):10882–94.
 - [21] Dai C, Shu Z, Li P, Mi J. Combustion characteristics of a methane jet flame in hot oxidant coflow diluted by h₂o versus the case by n₂. *Energy Fuels* 2018;32(1):875–88.
 - [22] Lee RC, Braden Wimmer Donn. Exhaust emission abatement by fuel variations to produce lean combustion. *SAE Trans* 1968:3025–44.
 - [23] Thawko A, Yadav H, Eyal A, Shapiro M, Tartakovsky L. Particle emissions of direct injection internal combustion engine fed with a hydrogen-rich reformate. *Int J Hydrogen Energy* 2019;44(52):28342–56.
 - [24] Pashchenko Dmitry. Energy optimization analysis of a thermochemical exhaust gas recuperation system of a gas turbine unit. *Energy Convers Manag* 2018;171:917–24.
 - [25] Popov SK, Svistunov IN, Garyaev AB, Serikov EA, Temyrkanova EK. The use of thermochemical recuperation in an industrial plant. *Energy* 2017;127:44–51.
 - [26] Gaber Christian, Demuth Martin, Prieler René, Schluckner Christoph, Hochenauer Christoph. An experimental study of a thermochemical regeneration waste heat recovery process using a reformer unit. *Energy* 2018;155:381–91.
 - [27] Upadhyay Mukesh, Lee Hyunjun, Kim Ayeon, Lee Sang-hun, Lim Hankwon. Cfd simulation of methane steam reforming in a membrane reactor: performance characteristics over range of operating window. *Int J Hydrogen Energy* 2021;46(59):30402–11.
 - [28] Chen Zhiqiang, Jiang Yong. Numerical investigation of the effects of h₂/co/syngas additions on laminar premixed combustion characteristics of nh₃/air flame. *Int J Hydrogen Energy* 2021;46(21):12016–30.
 - [29] Peters Norbert. *Turbulent combustion*. 2001.
 - [30] Lopez-Ruiz G, Fernandez-Akarregi AR, Diaz L, Urresti I, Alava I, Blanco JM. Numerical study of a laminar hydrogen diffusion flame based on the non-premixed finite-rate chemistry model; thermal nox assessment. *Int J Hydrogen Energy* 2019;44(36):20426–39.
 - [31] Habib Mohamed A, Mokheimer Esmail MA, Sanusi Sofihullahi Y, Nemitallah Medhat A. Numerical investigations of combustion and emissions of syngas as compared to methane in a 200 mw package boiler. *Energy Convers Manag* 2014;83:296–305.
 - [32] Gianfranco Scribano, Xinwei Cheng, and Manh-Vu Tran. Numerical simulation of the effects of hydrogen and carbon monoxide ratios on the combustion and emissions for syngas fuels in a radiant burner. *Energy*, 214: 118910.
 - [33] Asgari Nazli, Ahmed Sheikh F, I Farouk Tanvir, Padak Bihter. Nox formation in post-flame gases from syngas/air combustion at atmospheric pressure. *Int J Hydrogen Energy* 2017;42(38):24569–79.
 - [34] Lee Min Chul. Effects of h₂/co/ch₄ syngas composition variation on the no_x and co emission characteristics in a partially-premixed gas turbine combustor. *Sci China Technol Sci* 2016;59(12):1804–13.
 - [35] Alavandi SK, Agrawal AK. Experimental study of combustion of hydrogen–syngas/methane fuel mixtures in a porous burner. *Int J Hydrogen Energy* 2008;33(4):1407–15.
 - [36] Ghenai Chaouki. Combustion of syngas fuel in gas turbine can combustor. *Adv Mech Eng* 2010;2:342357.
 - [37] Giles Daniel E, Som Sibendu, Aggarwal Suresh K. Nox emission characteristics of counterflow syngas diffusion flames with airstream dilution. *Fuel* 2006;85(12–13):1729–42.
 - [38] Pashchenko Dmitry. Comparative analysis of hydrogen/air combustion cfd-modeling for 3d and 2d computational domain of micro-cylindrical combustor. *Int J Hydrogen Energy* 2017;42(49):29545–56.
 - [39] Pashchenko Dmitry. Hydrogen-rich fuel combustion in a swirling flame: cfd-modeling with experimental verification. *Int J Hydrogen Energy* 2020;45(38):19996–20003.
 - [40] Ansys Incorporated. *Fluent 2019 theory guide*. 2019.
 - [41] Mikulčić Hrvoje, Baleta Jakov, Wang Xuebin, Wang Jin, Qi Fengsheng, Wang Fan. Numerical simulation of ammonia/methane/air combustion using reduced chemical kinetics models. *Int J Hydrogen Energy* 2021;46(45):23548–63.
 - [42] Zhou Anqi, Xu Hongpeng, Xu Mingchen, Yu Wenbin, Li Zhenwei, Yang Wenming. Numerical investigation of biomass co-combustion with methane for nox reduction. *Energy* 2020;194:116868.
 - [43] Edward Launder Brian, Brian Spalding Dudley, et al. *Lectures in mathematical models of turbulence*. 1972.
 - [44] Yakhot Victor, Orszag Steven A. Renormalization group analysis of turbulence. i. basic theory. *J Sci Comput* 1986;1(1):3–51.
 - [45] Roman Centeno Felipe, Brittes Rogério, França Francis HR, Cristiano Vitorino da Silva. Application of the wsgg model for the calculation of gas–soot radiation in a turbulent non-premixed methane–air flame inside a cylindrical combustion chamber. *Int J Heat Mass Tran* 2016;93:742–53.
 - [46] Karyeyen Serhat. Combustion characteristics of a non-premixed methane flame in a generated burner under

- distributed combustion conditions: a numerical study. *Fuel* 2018;230:163–71.
- [47] Sivathanu YR, Faeth Gerard M. Generalized state relationships for scalar properties in non premixed hydrocarbon/air flames. *Combust Flame* 1990;82(2):211–30.
- [48] Jones WPa, Whitelaw JH. Calculation methods for reacting turbulent flows: a review. *Combust Flame* 1982;48:1–26.
- [49] Büyükkakın Mustafa Kemalettin, Öztuna Semiha. Numerical investigation on hydrogen-enriched methane combustion in a domestic back-pressure boiler and non-premixed burner system from flame structure and pollutants aspect. *Int J Hydrogen Energy* 2020;45(60):35246–56.
- [50] Hanson Ronald K, Salimian Siamak. Survey of rate constants in the n/h/o system. In: *Combustion chemistry*. Springer; 1984. p. 361–421.
- [51] Jenni AM Sidey, Mastorakos Epaminondas. Simulations of laminar non-premixed flames of methane with hot combustion products as oxidiser. *Combust Flame* 2016;163:1–11.
- [52] Nemitallah Medhat A, Rashwan Sherif S, Mansir Ibrahim B, Abdelhafez Ahmed A, Habib Mohamed A. Review of novel combustion techniques for clean power production in gas turbines. *Energy Fuels* 2018;32(2):979–1004.
- [53] Pascal Marquet. On the computation of moist-air specific thermal enthalpy. *Q J R Meteorol Soc* 2015;141(686):67–84.
- [54] Wischniewski Berndt. Air enthalpy calculator. 2021. Online, https://www.peacesoftware.de/einigewerte/luft_e.html. [Accessed 24 February 2021].
- [55] Gicquel Laurent YM, Gabriel Staffelbach, Poinso Thierry. Large eddy simulations of gaseous flames in gas turbine combustion chambers. *Prog Energy Combust Sci* 2012;38(6):782–817.
- [56] Glassman Irvin, Yetter Richard A, Glumac Nick G. *Combustion*. Academic press; 2014.
- [57] Kruse Stephan, Bruno Kerschgens, Berger Lukas, Varea Emilien, Pitsch Heinz. Experimental and numerical study of mild combustion for gas turbine applications. *Appl Energy* 2015;148:456–65.
- [58] Mokheimer Esmail MA, Sanusi Yinka S, Habib Mohamed A. Numerical study of hydrogen-enriched methane–air combustion under ultra-lean conditions. *Int J Energy Res* 2016;40(6):743–62.
- [59] Schluckner C, Gaber C, Landfahner M, Demuth M, Hochenauer C. Fast and accurate cfd-model for nox emission prediction during oxy-fuel combustion of natural gas using detailed chemical kinetics. *Fuel* 2020;264:116841.
- [60] Escudero Ana I, Aznar María, Díez Luis I, Mayoral M Carmen, José M Andrés. From o₂/co₂ to o₂/h₂o combustion: the effect of large steam addition on anthracite ignition, burnout and nox formation. *Fuel Process Technol* 2020;206:106432.
- [61] Pugh DG, Bowen PJ, Marsh Richard, Crayford AP, Runyon Jon, Morris Steven, Valera-Medina A, Giles Anthony. Dissociative influence of h₂o vapour/spray on lean blowoff and nox reduction for heavily carbonaceous syngas swirling flames. *Combust Flame* 2017;177:37–48.